

SHAMKUMAR A

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SUMMARY

Passionate ECE student with hands-on experience in Embedded Systems and IoT development using ESP32, Embedded C, and MQTT protocol. Developed a Smart Classroom Automation System for real-time wireless monitoring and control of multiple devices. Skilled in PCB design using EasyEDA and IoT cloud platforms. Seeking a hands-on internship in Embedded Systems and IoT.

EDUCATION

B.E. – Electronics and Communication Engineering 2023 – 2027
Government College of Engineering, Bargur | CGPA: 7.75
Higher Secondary Education (HSC) 2020 – 2023
Sri Vidya Mandir Senior Secondary School | HSC: 85.5% | SSLC: 75.8%

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Embedded C
Hardware & Platforms: ESP32, Arduino, NodeMCU, Relay Modules, Sensors
Communication Protocols: MQTT, I2C, SPI, UART
Tools & Software: Proteus, MPLAB, Wokwi, EasyEDA, Blynk, Arduino IDE
Core Knowledge: Digital Electronics, Embedded Systems, IoT, PCB Design

PROJECTS

Smart Classroom Automation System Sep 2025
Technologies: ESP32, MQTT, Relay Modules, Sensors, Blynk, EasyEDA

- Developed an IoT-based automation system using ESP32 and MQTT for wireless control of 6+ classroom devices including lights, fans, and projectors.
- Implemented automatic sensor-triggered control for motion detection, ambient light, and temperature monitoring.
- Designed a modular relay PCB layout in EasyEDA, reducing wiring complexity and enabling easy scalability of additional devices.
- Integrated Blynk cloud dashboard for real-time monitoring and remote control via mobile application.

Temperature & Humidity Monitor College Lab Project
Technologies: Arduino, DHT11 Sensor, LCD Display, Embedded C

- Built a real-time monitoring system using DHT11 sensor and Arduino, displaying live data on a 16x2 LCD screen.

WORKSHOPS ATTENDED

- IoT Based Design and Implementation Using Modern Microcontrollers
- Electronic Assembly and Electrical Measurement Using Multimeter and Clamp Meter

ACHIEVEMENTS

- GATE 2026 – Appeared (ECE) | Currently Preparing for GATE 2027

SOFT SKILLS

Problem Solving | Business Communication | Effective Presentation | Team Collaboration